Data Visual

Date	26 Feb 2026
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Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	3 Marks

Section 1: Data Preprocessing Steps

S.N	Preprocessing Details
1	**Dataset: EVIndia.csv** **Step 1 - Rename Columns** • "PriceRange(Lakhs)" → Price (Lakhs) • "BootSpace" → Boot Space (L) • "VehicleType" → Vehicle Type **Step 2 - Change Data Types** • Price (Lakhs) → Number (Decimal) • Car, Style, Transmission, Vehicle Type, Capacity, BaseModel, TopModel → String • Range → String (will be parsed by Calculated Field) **Step 3 - Calculated Field: Range (km)** Extracts numeric km value. Example: "312 Km/Full Charge" → 312 Formula: INT(LEFT([Range], FIND([Range], " ") - 1)) **Step 4 - Calculated Field: Boot Space (Litres)** Handles 'na' string as NULL. Example: "350 L" → 350 Formula: IF [Boot Space]="na" THEN NULL ELSE INT(LEFT([Boot Space (L)], FIND([Boot Space (L)], " ") - 1)) END
2	**Dataset: PriceEuro.csv & Germany.csv** **Step 1 - Rename Columns** • Standardize column names across both datasets • Map price fields to "Price (EUR)" and "Price (GBP)" as applicable • Rename "BootSpace" → Boot Space (L) and "VehicleType" → Vehicle Type **Step 2 - Change Data Types** • Price fields (Euro/GBP) → Number (Decimal) • Car, Style, Transmission, Vehicle Type, Capacity, BaseModel, TopModel → String • Range → String (will be parsed by Calculated Field) **Step 3 - Calculated Field: Range (km)**

	Extracts numeric km value Formula: INT(LEFT([Range], FIND([Range], " ") - 1)) **Step 4 - Calculated Field: Boot Space (Litres)** Handles 'na' or missing values as NULL Formula: IF [Boot Space]="na" THEN NULL ELSE INT(LEFT([Boot Space (L)], FIND([Boot Space (L)], " ") - 1)) END
3	**Calculated Fields Summary (All Datasets)** **Field: Seats** Extracts integer seat count from Capacity field Formula: INT(LEFT(TRIM([Capacity]), 1)) **Field: Brand** Extracts manufacturer name from Car field Formula: LEFT([Car], FIND([Car], " ") - 1) **Field: Range (km)** Extracts numeric range from string data Formula: INT(LEFT([Range], FIND([Range], " ") - 1)) **Field: Boot Space (Litres)** Converts string to numeric, handles NULL values Formula: IF [Boot Space]="na" THEN NULL ELSE INT(LEFT([Boot Space (L)], FIND([Boot Space (L)], " ") - 1)) END

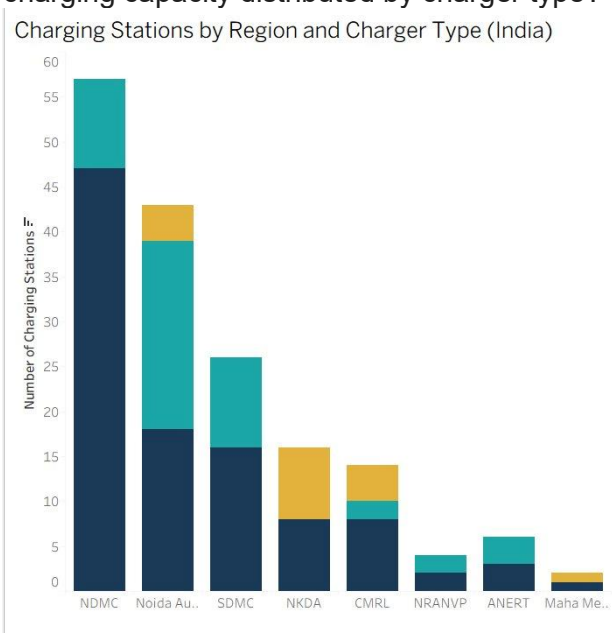
Key Preprocessing Notes:

- All datasets follow standardized column naming and data type conversion
- Calculated fields use consistent Tableau formulas across all three datasets
- String-to-numeric conversion ensures proper visualization and range analysis
- NULL handling allows for accurate aggregations with missing data
- Cross-currency support (Lakhs, EUR, GBP) enables international comparison

Section 2: Business Questions with Visualizations

1. Charging Station Distribution Across India

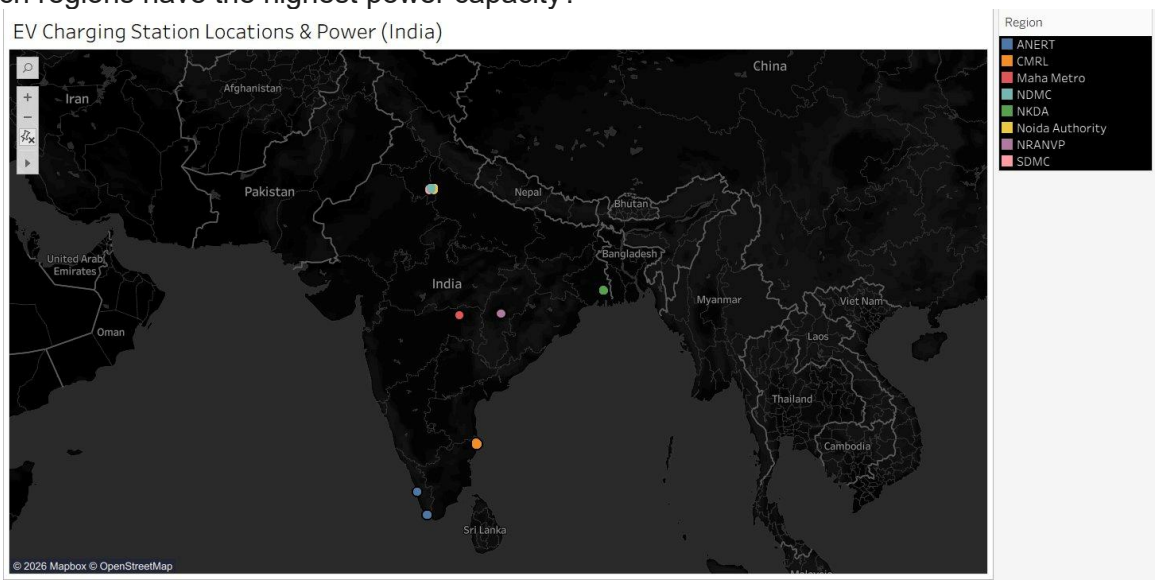
Business Question: Which regions in India have the most EV charging infrastructure, and how is the charging capacity distributed by charger type?



Insight: NDMC and Noida Authority regions dominate with 57 and 42 charging stations respectively. The presence of both fast and regular chargers indicates varying infrastructure maturity across regions.

2. Geographic Distribution of EV Charging Stations

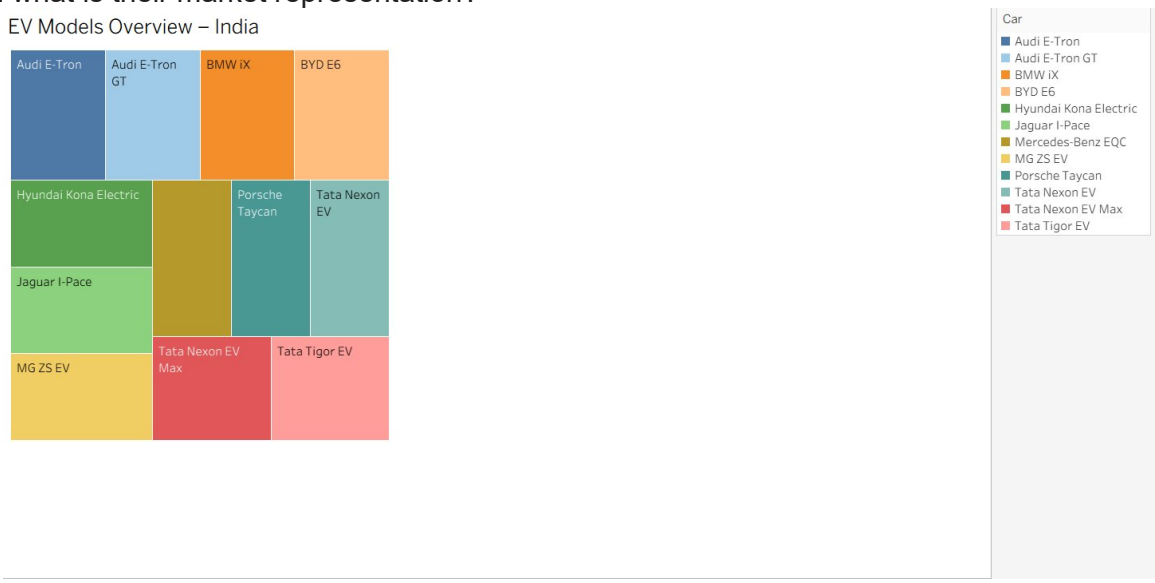
Business Question: What is the geographic coverage of EV charging stations across India, and which regions have the highest power capacity?



Insight: Station distribution shows concentration in major metro areas with sparse coverage in tier 2 cities. This reveals the critical gap in charging infrastructure for long-distance EV travel.

3. Available EV Models in India Market

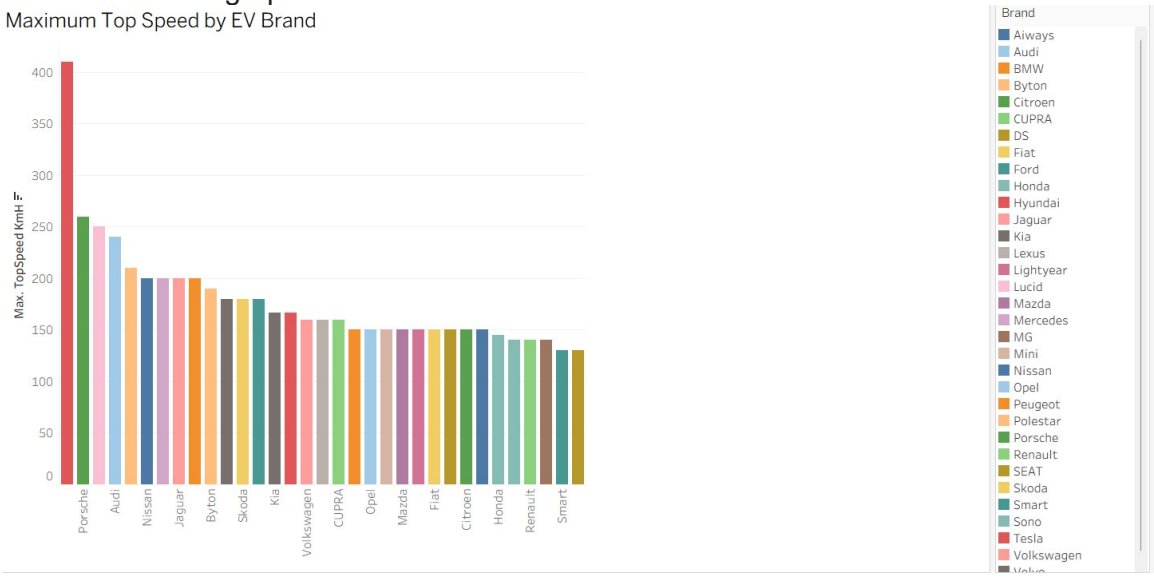
Business Question: Which EV brands and models are currently available in the Indian market, and what is their market representation?



Insight: Market is dominated by premium brands (Audi, BMW) with Tata and Hyundai providing affordable options. The portfolio shows a premium-focused market with limited budget segment options.

4. Performance Comparison: Maximum Top Speed by Brand

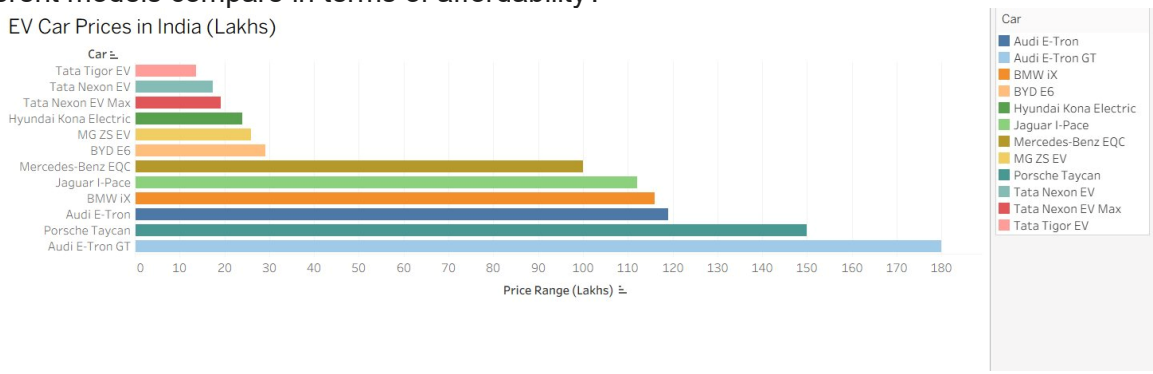
Business Question: How do different EV brands compare in terms of maximum top speed, and which brands offer high-performance vehicles?



Insight: Porsche leads with 415 km/h, followed by luxury brands like Audi and Tesla. Performance variance indicates differentiated positioning from economy to luxury segments.

5. Pricing Analysis: EV Car Prices in India

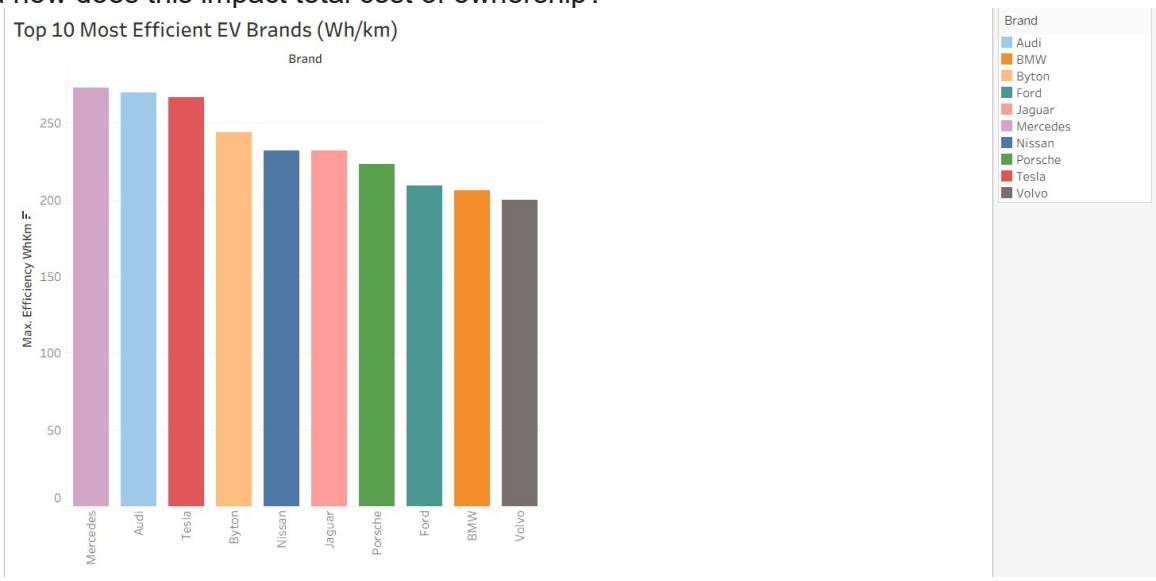
Business Question: What is the price range distribution of EV vehicles in India, and how do different models compare in terms of affordability?



Insight: Price range spans from ₹13L (Tata Tigor EV) to ₹180L (Audi E-Tron GT). Clear market segmentation shows entry-level, mid-range, and premium positioning.

6. Energy Efficiency: Top 10 Most Efficient EV Brands

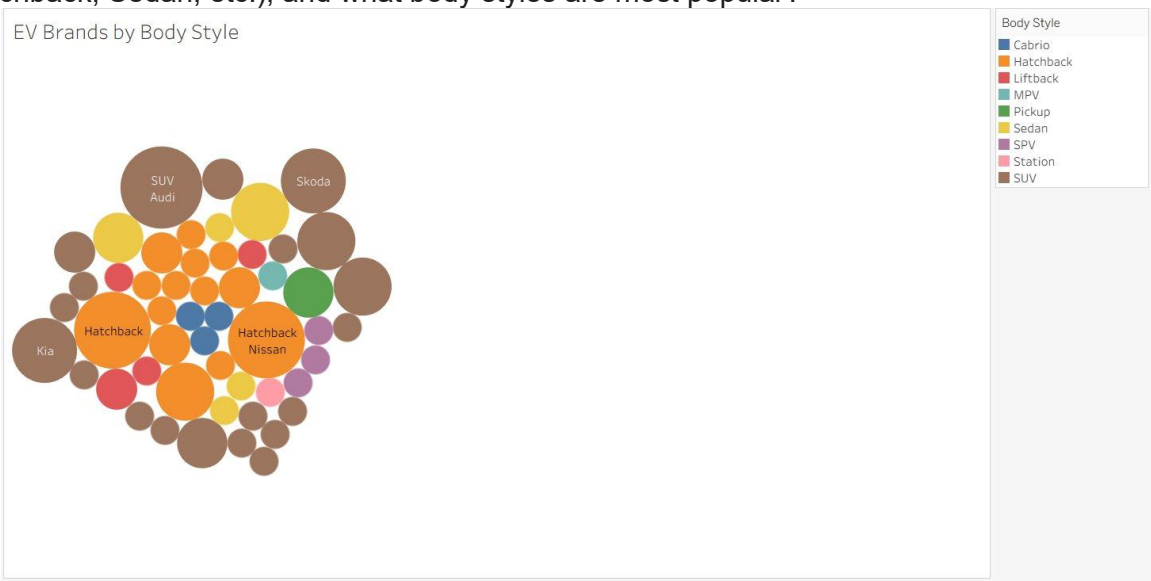
Business Question: Which EV brands are most energy efficient in terms of Wh/km consumption, and how does this impact total cost of ownership?



Insight: Mercedes-Benz, Audi, and Tesla are most efficient at ~270 Wh/km. Higher efficiency reduces charging frequency and operational costs significantly.

7. Market Segmentation: EV Brands by Body Style

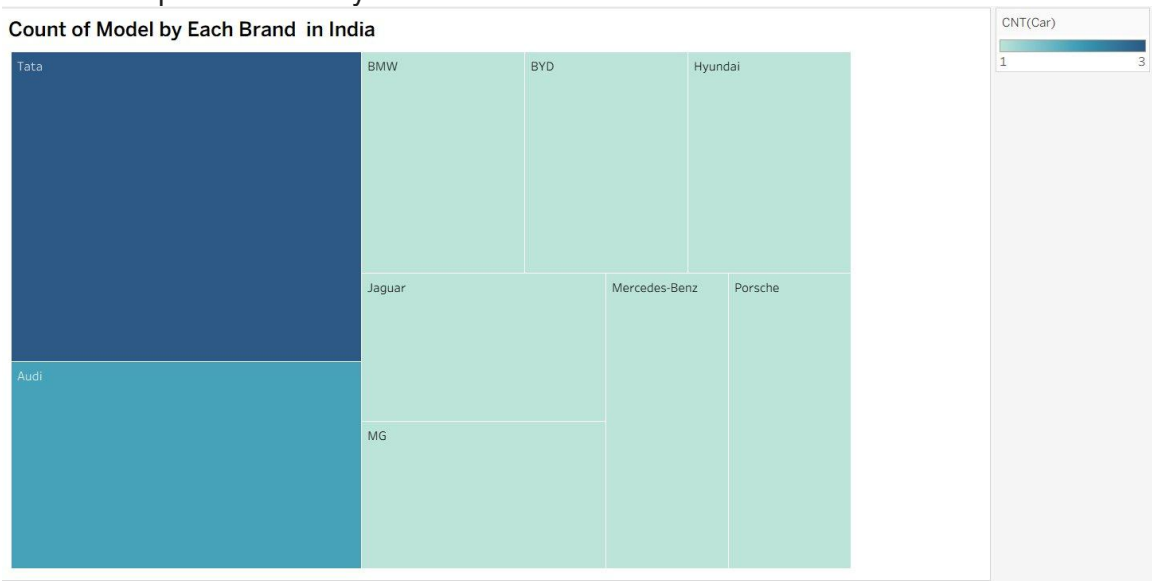
Business Question: How are different EV brands distributed across body styles (SUV, Hatchback, Sedan, etc.), and what body styles are most popular?



Insight: SUV segment dominated by Audi and Skoda, while Hatchback represents mass-market offerings (Nissan, Kia). SUV preference aligns with Indian consumer trends for spacious vehicles.

8. Market Dominance: Count of Models by Brand in India

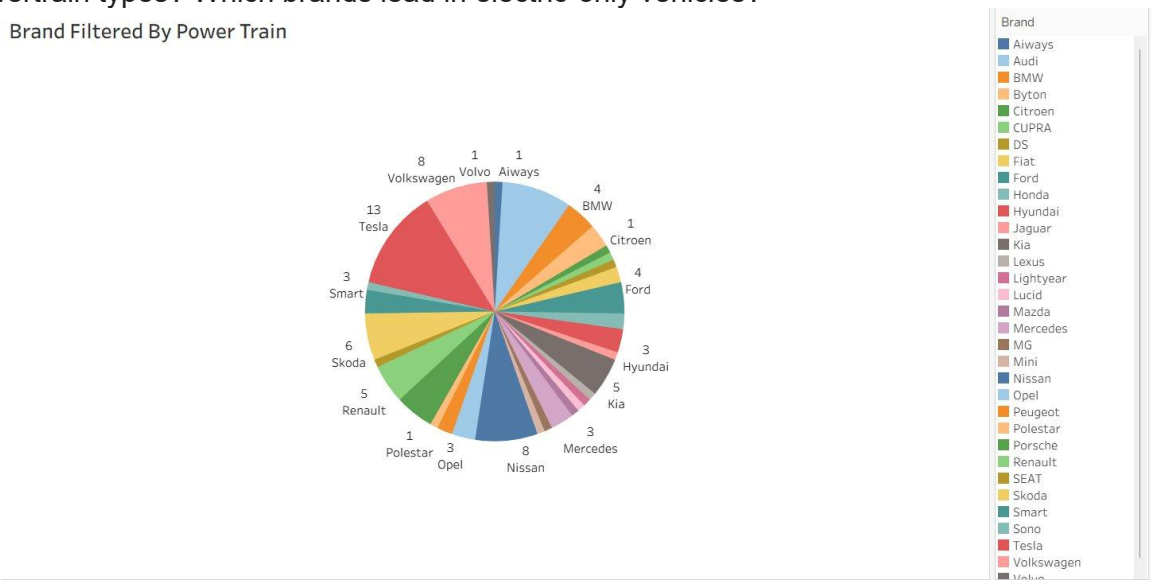
Business Question: Which brands have the highest number of models available in India, and what is the competitive intensity in the market?



Insight: Tata dominates with the most models available, indicating strong market commitment. Audi and BMW follow with limited but premium portfolio, showing selective market strategy.

9. Market Share: Brand Distribution (Power Train Filtered)

Business Question: What is the market share of different EV brands when filtering by specific powertrain types? Which brands lead in electric-only vehicles?



Insight: Tesla leads with 13 models on powertrain filter, followed by Nissan (8 models) and Hyundai (3 models). This shows market consolidation with Tesla and traditional automakers competing for EV dominance.

Key Findings & Recommendations

- **Infrastructure Gap:** While major metros have adequate charging stations, tier 2 and tier 3 cities require urgent investment for mass EV adoption.
- **Premium Market Focus:** Current offerings skew towards premium segment (Audi, BMW, Mercedes). Budget EV options need expansion for mass-market penetration.
- **Efficiency Leaders:** Mercedes-Benz and Audi vehicles lead in efficiency, directly impacting customer operational costs and charging requirements.
- **Brand Diversification:** Tata's multi-model approach contrasts with selective premium strategies, enabling broader market reach.
- **Body Style Preference:** SUVs dominate, reflecting consumer preference for spacious vehicles, guiding manufacturer product development.